1) $t_0 = 0m\ T_0 = 90°F$
$t_{60} = 1h\ T_{60} = 87°F$
$T_{Room} = 70°$

$$\frac{dT}{dt} = C(T - T_{room})$$

$$\frac{dT}{dt} = C(T - 70)$$

$$dT/T-70 = C\,dt$$

$$\int dT/T-70 = \int C\,dt$$

$$\ln(T-70) = Ct + K$$

$$T-70 = e^{Ct+K}$$

$$T = 70 + e^{Ct+}$$

$$T = 70 + e^{K}e^{Ct}$$

$$T = 70 + Ae^{Ct},\ A = e^{K}$$

$$T(0) = 90 = 70 + Ae^{C(0)}$$

$$90 = 70 + A(1)$$

$$A = 20$$

$$T(1) = 87 = 70 + 20e^{C(1)}$$

$$17 = 20e^{C}$$

$$\ln(17/20) = C$$

$$-0.1625 = C$$

$$T(t) = 70 + 20e^{-0.1625t}$$

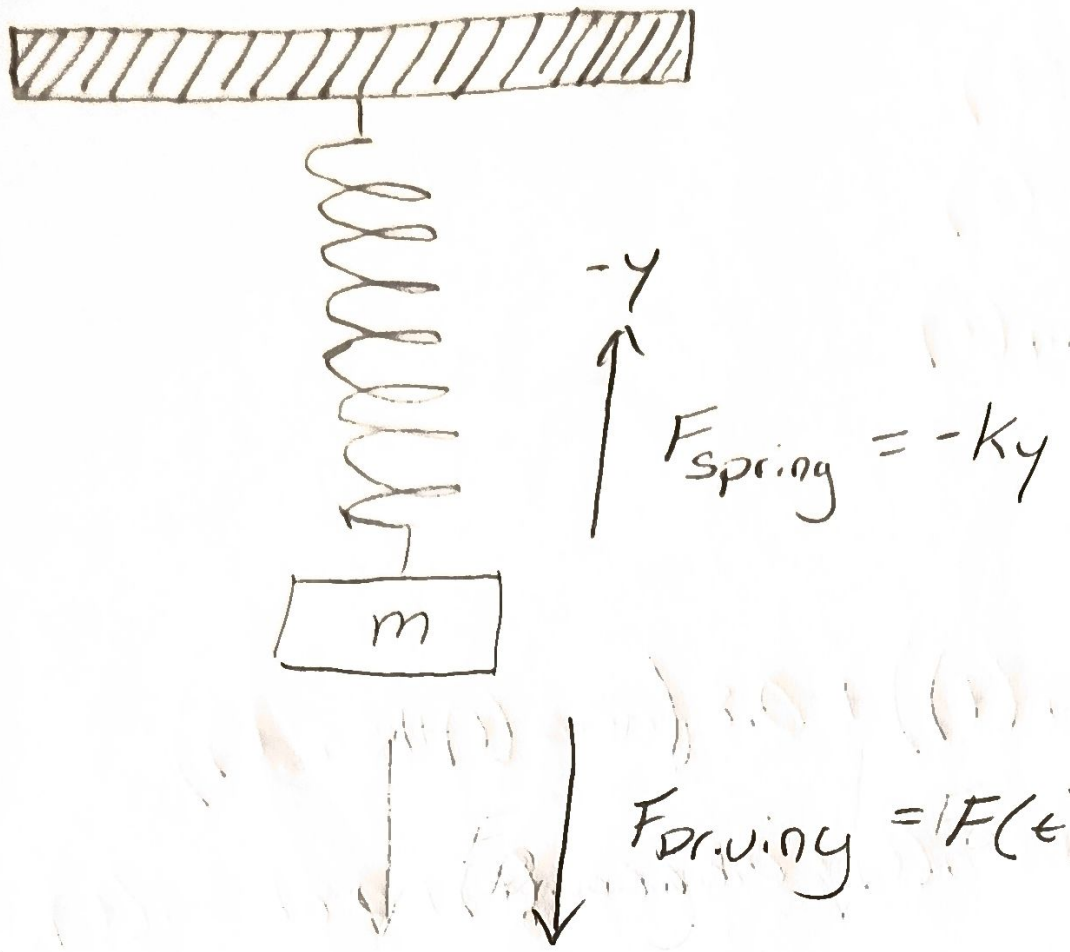
$$98.6° = 70 + 20e^{-0.1625t}$$

$$28.6° = 20e^{-0.1625t}$$

$$\ln\left(\frac{28.6°}{20}\right) = -0.1625t$$

$t = -2.201$ hours or
$\approx$ 8:18am

2|



$\sum F = my''$ $+y$

$-ky + F(t) = my''$

$F(t) = my'' + ky$

$0.3y'' + 0.8y = \cos(t) + 4\sin(t)$

$m = 0.3\,kg$

$F(t) = [\cos(t) - 4\sin(t)]N$

$k = 0.8\,N/m$

$y(0) = 0m$

Homogeneous Solution:

$y_h(t) = A\cos(w_o t) + B\sin(w_o t)$,

$w_o = \sqrt{\frac{k}{m}} = \sqrt{\frac{0.8}{0.3}} = 1.632$ — From the book

$y_h(t) = A\cos(1.632t) + B\sin(1.632t)$

2| Particular Solution

$$y_p(t) = a\cos(t) + \sin(t)$$
$$y_p'(t) = -a\sin(t) + b\cos(t)$$
$$y_p''(t) = -a\cos(t) - b\sin(t)$$

$$F(t) = 0.3y'' + 0.8y$$
$$= 0.3(a\cos(t) - b\sin(t)) + 0.8(a\cos(t) + b\sin(t))$$
$$= -0.3a\cos(t) - 0.3b\sin(t) + 0.8a\cos(t) + 0.8b\sin(t)$$
$$= (-0.3a + 0.8a)\cos(t) + (-0.3b + 0.8b)\sin(t)$$
$$= 0.5a\cos(t) + 0.5b\sin(t)$$

$$0.5a\cos(t) + 0.5b\sin(t) = \cos(t) - 4\sin(t)$$

$0.5a = 1$ $\quad$ $0.5b = -4$

$a = 2$ $\quad$ $b = -8$

$$y_p(t) = 2\cos(t) + 8\sin(t)$$

$$y(t) = A\cos(1.632t) + B\sin(1.632t) + 2\cos(t) - 8\sin(t)$$
$$y(0) = 0 = A + 2 \rightarrow A = -2$$

$$\boxed{y(t) = -2\cos(1.632t) + B\sin(1.632t) + 2\cos(t) - 8\sin(t)}$$

*Not sure what the 2nd value to find B would be

3)

$$A = \begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 8 & 7 \\ 9 & 15/2 & 6 \\ 9 & 6 & 3 \end{bmatrix}$$

$$\equiv \begin{bmatrix} 9 & 8 & 7 \\ 0 & -1/2 & -1 \\ 0 & -2 & -4 \end{bmatrix}$$

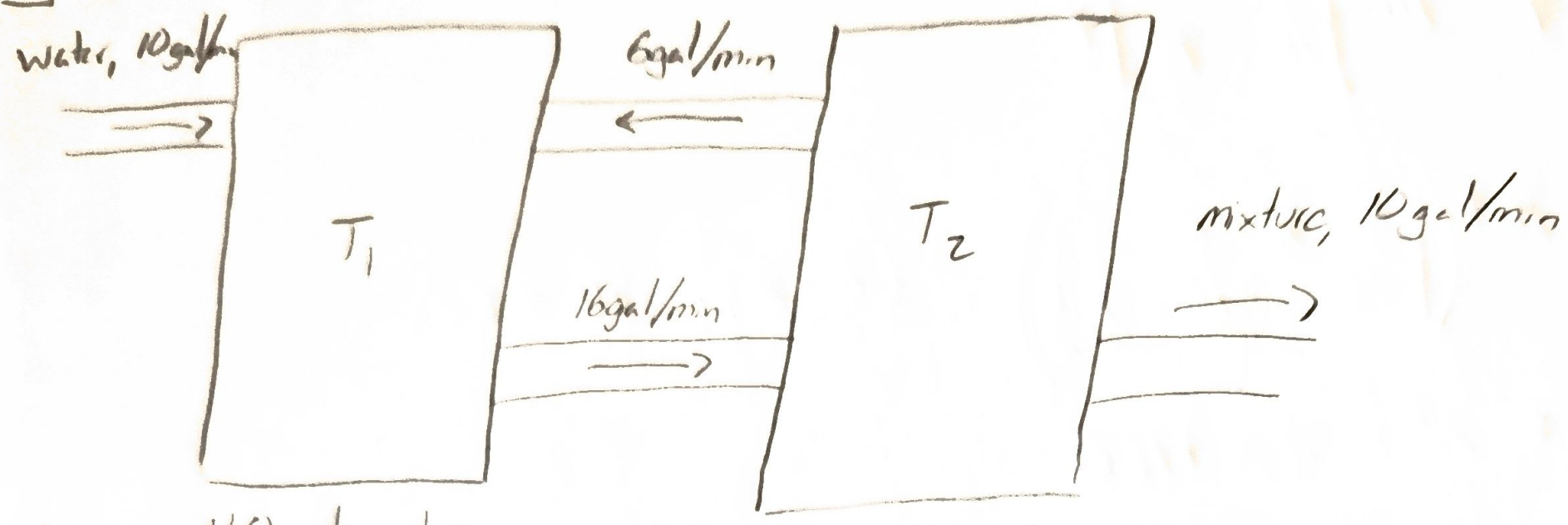
$$= \begin{bmatrix} 9 & 8 & 7 \\ 0 & 1 & 2 \\ 0 & -2 & -4 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 8 & 7 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 8/9 & 7/9 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

A has rank 2 and does not have full rank

4)



110 gal water
150 lbs salt
at $t=0$

In: 16 gal/min
Out: 16 gal/min

110 gal water
0 lbs salt
at $t=0$

In: 16 gal/min
Out: 16 gal/min

volume is constant

Salt rate = concentration * flow rate

$= \frac{lbs}{110 gal} * \frac{gal}{min} = lbs/min$

$concentration = \frac{mass}{volume}$

$y_1' = C_1 y_1 + C_2 y_2$

$y_1' = \frac{6}{110} y_2 - \frac{16}{110} y_1$

$y_2' = C_1 y_2 + C_2 y_2$

$y_2' = \frac{16}{110} y_1 - \frac{16}{110} y_2$

$$y' = \frac{1}{110} \begin{bmatrix} -16 & 6 \\ 16 & -16 \end{bmatrix} y$$

4)

$$Y' = \frac{1}{110}\begin{bmatrix} -16 & 6 \\ 16 & -16 \end{bmatrix} Y$$

$$\det\left(\begin{bmatrix} -16-\lambda & 6 \\ 16 & -16-\lambda \end{bmatrix}\right) = (-16-\lambda)^2 - 96 = 0$$

$$= \lambda^2 + 32\lambda + 160$$

$$\lambda = \frac{-32 \pm \sqrt{32^2 - 4(1)(160)}}{2(1)}$$

$$\lambda = -16 \pm 4\sqrt{6}$$

$$(A - \lambda I) = 0$$

$$= \begin{bmatrix} -16-(-16+4\sqrt{6}) & 6 \\ 16 & -16-(-16+4\sqrt{6}) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 4\sqrt{6} & 6 \\ 16 & 4\sqrt{6} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$= 4\sqrt{6}\,x_1 + 6x_2 = 0 \Rightarrow x_2 = \frac{4\sqrt{6}}{6}x_1 = \frac{2\sqrt{6}}{3}x_1$$

Similarly for $\lambda = -16 - 4\sqrt{6}$, $x_2 = -\frac{2\sqrt{6}}{3}x_1$

$x_1 = 3 \Rightarrow \begin{bmatrix} 3 \\ 2\sqrt{6} \end{bmatrix}, \begin{bmatrix} 3 \\ -2\sqrt{6} \end{bmatrix}$

4)
$$\begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} = C_1 \begin{bmatrix} 3 \\ 2\sqrt{6} \end{bmatrix} e^{(-4+4\sqrt{6})t} + C_2 \begin{bmatrix} 3 \\ -2\sqrt{6} \end{bmatrix} e^{(-4-4\sqrt{6})t}$$

Tank 1 initially has 150 lbs of salt at $t=0$

$$Y_1(0) = 150$$

Tank 0 has 0 lbs

$$Y_2(0) = 0$$

$$Y_1 = 3C_1 e^{(-4+4\sqrt{6})t} + 3C_2 e^{(-4-4\sqrt{6})t}$$

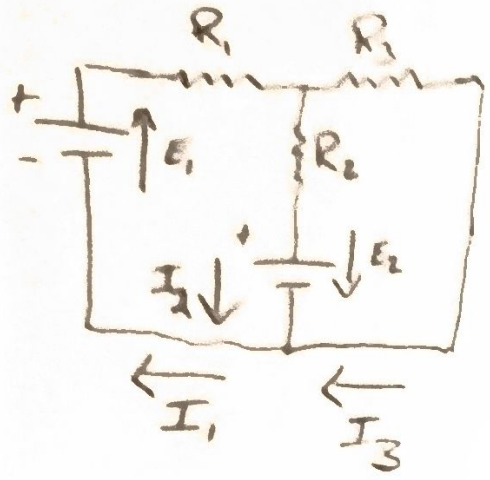
$$Y_2 = 2\sqrt{6}C_1 e^{(-4+4\sqrt{6})t} - 2\sqrt{6}C_2 e^{(-4-4\sqrt{6})t}$$

$$Y_1(0) = 150 = 3C_1 + 3C_2 \qquad C_1 + C_2 = 50$$
$$Y_2(0) = 0 = 2\sqrt{6}C_1 - 2\sqrt{6}C_2 \Rightarrow C_1 = C_2$$
$$C_1, C_2 = 25$$

$$\boxed{Y_1(t) = 75\left(e^{(-4+4\sqrt{6})t} + e^{(-4-4\sqrt{6})t}\right)}$$

$$\boxed{Y_2(t) = 50\sqrt{6}\left(e^{(-4+4\sqrt{6})t} - e^{(-4-4\sqrt{6})t}\right)}$$

5)



$$-I_1 + I_2 + I_3 = 0$$
$$R_1 I_1 + R_2 I_2 = E_1 + E_2$$
$$R_2 I_2 - R_3 I_3 = E_2$$

$R_1 = 8\Omega \quad R_2 = 4\Omega \quad R_3 = 12\Omega \quad E_1 = 12V \quad E_2 = 24V$

$$-I_1 + I_2 + I_3 = 0$$
$$8I_1 + 4I_2 = 36$$
$$4I_2 - 12I_3 = 24$$

$$\Rightarrow \left[\begin{array}{ccc|c} -1 & 1 & 1 & 0 \\ 8 & 4 & 0 & 36 \\ 0 & 4 & -12 & 24 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & -1 & -1 & 0 \\ 8 & 4 & 0 & 36 \\ 0 & 4 & -12 & 24 \end{array}\right] = \left[\begin{array}{ccc|c} 1 & 0 & -1/3 & 3 \\ 0 & 1 & 2/3 & 3 \\ 0 & 0 & -11/3 & 3 \end{array}\right] =$$

$$\left[\begin{array}{ccc|c} 1 & -1 & -1 & 0 \\ 0 & 12 & 8 & 36 \\ 0 & 4 & -12 & 24 \end{array}\right] = \left[\begin{array}{ccc|c} 1 & 0 & -1/3 & 3 \\ 0 & 1 & 2/3 & 3 \\ 0 & 0 & 1 & -9/11 \end{array}\right] =$$

$$\left[\begin{array}{ccc|c} 1 & -1 & -1 & 0 \\ 0 & 1 & 2/3 & 3 \\ 0 & 1 & -3 & 6 \end{array}\right] = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 30/11 \\ 0 & 1 & 0 & 39/11 \\ 0 & 0 & 1 & -9/11 \end{array}\right]$$

$$\boxed{I_1 = 30/11\ A,\ I_2 = 39/11\ A,\ I_3 = -9/11\ A}$$

6)

$$B=\begin{bmatrix} -1 & 3 & 0 \\ -7 & 2 & 1 \\ -2 & 4 & -3 \end{bmatrix}$$

B^{-1}

$$\left[\begin{array}{ccc|ccc} -1 & 3 & 0 & 1 & 0 & 0 \\ -7 & 2 & 1 & 0 & 1 & 0 \\ -2 & 4 & -3 & 0 & 0 & 1 \end{array}\right] =$$

$$\left[\begin{array}{ccc|ccc} 1 & -3 & 0 & -1 & 0 & 0 \\ -7 & 2 & 1 & 0 & 1 & 0 \\ -2 & 4 & -3 & 0 & 0 & 1 \end{array}\right] =$$

$$\left[\begin{array}{ccc|ccc} 1 & -3 & 0 & -1 & 0 & 0 \\ 0 & -19 & 1 & -7 & 1 & 0 \\ 0 & -2 & -3 & -2 & 0 & 1 \end{array}\right] =$$

$$\left[\begin{array}{ccc|ccc} 1 & -3 & 0 & -1 & 0 & 0 \\ 0 & 1 & -1/19 & 17/19 & -1/19 & 0 \\ 0 & -2 & -3 & -2 & 0 & 1 \end{array}\right] =$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -3/19 & 2/19 & -3/19 & 0 \\ 0 & 1 & -1/19 & 7/19 & -1/19 & 0 \\ 0 & 0 & -59/19 & -24/19 & -2/19 & 1 \end{array}\right] =$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -3/19 & 2/19 & -3/19 & 0 \\ 0 & 1 & -1/19 & 7/19 & -1/19 & 0 \\ 0 & 0 & 1 & 24/59 & 2/59 & -19/59 \end{array}\right] =$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 10/59 & -9/59 & -3/59 \\ 0 & 1 & 0 & 23/59 & -3/59 & -1/59 \\ 0 & 0 & 1 & 24/59 & 2/59 & -19/59 \end{array}\right] =$$

$$\boxed{B^{-1} = \begin{bmatrix} 10/59 & -9/59 & -3/59 \\ 23/59 & -3/59 & -1/59 \\ 24/59 & 2/59 & -19/59 \end{bmatrix}}$$

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6)

$$B = \begin{bmatrix} -1 & 3 & 0 \\ -7 & 2 & 1 \\ -2 & 4 & -3 \end{bmatrix} \qquad B^{-1} = \frac{1}{59}\begin{bmatrix} 10 & -9 & -3 \\ 23 & -3 & -1 \\ 29 & 2 & -19 \end{bmatrix}$$

$$BB^{-1} = \begin{bmatrix} -1 & 3 & 0 \\ -7 & 2 & 1 \\ -2 & 4 & -3 \end{bmatrix}\begin{bmatrix} 10 & -9 & -3 \\ 23 & -3 & -1 \\ 29 & 2 & -19 \end{bmatrix}$$

$$= \begin{bmatrix} -1(10)+3(23)+0(24) & -1(-9)+3(-3)+0(2) & -1(-3)+3(-1)+0(-19) \\ -7(10)+2(23)+1(24) & -7(-9)+2(-3)+1(2) & -7(-3)+2(-1)+1(-19) \\ -2(10)+4(23)+-3(24) & -2(-9)+4(-3)+-3(2) & -2(-3)+4(-1)+-3(-19) \end{bmatrix}$$

$$= \frac{1}{59}\begin{bmatrix} 59 & 0 & 0 \\ 0 & 59 & 0 \\ 0 & 0 & 59 \end{bmatrix}$$

$$BB^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$B^{-1}B = \begin{bmatrix} 10 & -9 & -3 \\ 23 & -3 & -1 \\ 29 & 2 & -19 \end{bmatrix}\begin{bmatrix} -1 & 3 & 0 \\ -7 & 2 & 1 \\ -2 & 4 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 10(-1)+-9(-7)+-3(-2) & 10(3)+-9(2)+-3(4) & 10(0)+-9(1)+-3(-3) \\ 23(-1)+-3(-7)+-1(-2) & 23(3)+-3(2)+-1(4) & 23(0)+-3(1)+-1(-3) \\ 29(-1)+2(-7)+-19(-2) & 29(3)+2(2)+-19(4) & 29(0)+2(1)+-19(-3) \end{bmatrix}$$

$$B^{-1}B = \frac{1}{59}\begin{bmatrix} 59 & 0 & 0 \\ 0 & 59 & 0 \\ 0 & 0 & 59 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = BB^{-1} = \underline{I}$$

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7] $C = \begin{bmatrix} 7 & 2 & -1 \\ 2 & 7 & 1 \\ -1 & 1 & 8.5 \end{bmatrix}$

$$\det(C - \lambda I) = \det\left(\begin{bmatrix} 7-\lambda & 2 & -1 \\ 2 & 7-\lambda & 1 \\ -1 & 1 & 8.5-\lambda \end{bmatrix}\right) =$$

$$= (7-\lambda)\left((7-\lambda)(8.5-\lambda)-1\right) - 2\left(2(8.5-\lambda)+1\right) + -1\left(2+(7-\lambda)\right)$$

$$= (7-\lambda)(59.5 - 7\lambda - 8.5\lambda + \lambda^2 - 1) - 2(17 - 2\lambda + 1) - (9 - \lambda)$$

$$= (7-\lambda)(58.5 - 15.5\lambda + \lambda^2) + 4\lambda - 34 + 9 + \lambda$$

$$= 409.5 - 108.5\lambda + 7\lambda^2 - 58.5\lambda + 15.5\lambda^2 - \lambda^3 + 5\lambda - 25$$

$$= -\lambda^3 + 22.5\lambda^2 - 162\lambda + 364.5$$

Gotta throw in the towel here.
But to complete 7 I would find the roots of that polynomial to find the eigenvalues. Then I would find x in $(A - \lambda I)x = 0$ to get the eigenvectors.